END-TO-END DATA ANALYTICS

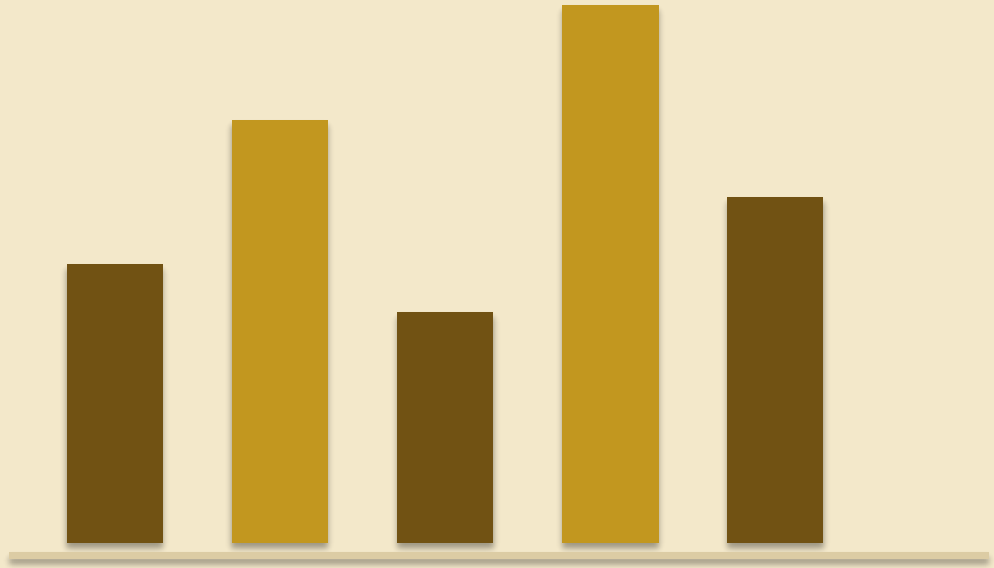
Tech Products Sales Analysis

Analyzing 1,000 sales transactions to understand revenue, product performance, sales channels and regional patterns.

PYTHON • MYSQL • POWER BI

PROJECT PRESENTATION

SALES PERFORMANCE



₹51.20M

Total sales from the analyzed dataset

DATASET OVERVIEW

A transaction-level view of technology product sales across India

The notebook shows 1,000 records and 11 fields covering products, quantities, unit prices, dates, customers, states, channels, payment types and sales.

1,000

Sales records

10

Unique products

28

States represented

Date range in the sample spans 2023–2024; 334 distinct date values were observed.

From raw spreadsheet to analysis-ready data

The uploaded notebook documents the Python workflow and checks.

Load data

Read the Excel file into a pandas DataFrame.

Explore

Review sample rows, data types and descriptive statistics.

Quality checks

Notebook reports zero missing values and no duplicated rows.

Standardize

Replace spaces in column names and rename Unit_Price.

Load to MySQL

Write 1,000 records to the salesdata table.

SALES SNAPSHOT

Key metrics from the dataset

Sales and quantity totals are calculated from the notebook's 1,000-row descriptive output.

TOTAL SALES

₹51,202,117

Sum of the Sales column

TOTAL QUANTITY

4,899

Total units across transactions

TRANSACTIONS

1,000

Distinct transaction IDs

Average sales per row: ₹51,202.12 • Average quantity per row: 4.899

Compare products by sales and quantity

The SQL script groups by product and sorts by total sales; product-level totals are not printed in the uploaded notebook output.

PRODUCT VARIETY

10 unique products appear in the dataset.

MOST FREQUENT PRODUCT

Printer appears 113 times in the notebook's descriptive summary.

SQL ANALYSIS

GROUP BY Product
SUM(Quantity)
SUM(Sales)
ORDER BY total_sales DESC

This ranks products by revenue once the query is executed.

Compare online, store and direct sales

The dataset contains three sales-channel categories; the SQL script calculates transactions and revenue by channel.

CHANNEL CATEGORIES

Online • Store • Direct

SQL METRICS

Transaction count and total sales by channel.

CHANNEL COMPARISON

Group by Sales_Channel
Count transactions
Sum sales
Sort by total sales

The notebook summary reports Store as the most frequent channel, with 346 rows.

SALES TREND

Track sales over time

Use Date on the X-axis and Sales on the Y-axis in Power BI; aggregate by month for a clearer trend.



Replace the illustrative line with the Power BI chart based on the actual data.

DATE COVERAGE

2023–2024

334 distinct dates are present. This illustration is not the actual daily sales series.

Find important markets and high-value customers

The SQL script includes state-wise sales and a top-five customer spending query.

STATE-WISE SALES

Aggregate Sales by State and sort from highest to lowest.

TOP FIVE CUSTOMERS

Rank customers by total spending and order count.

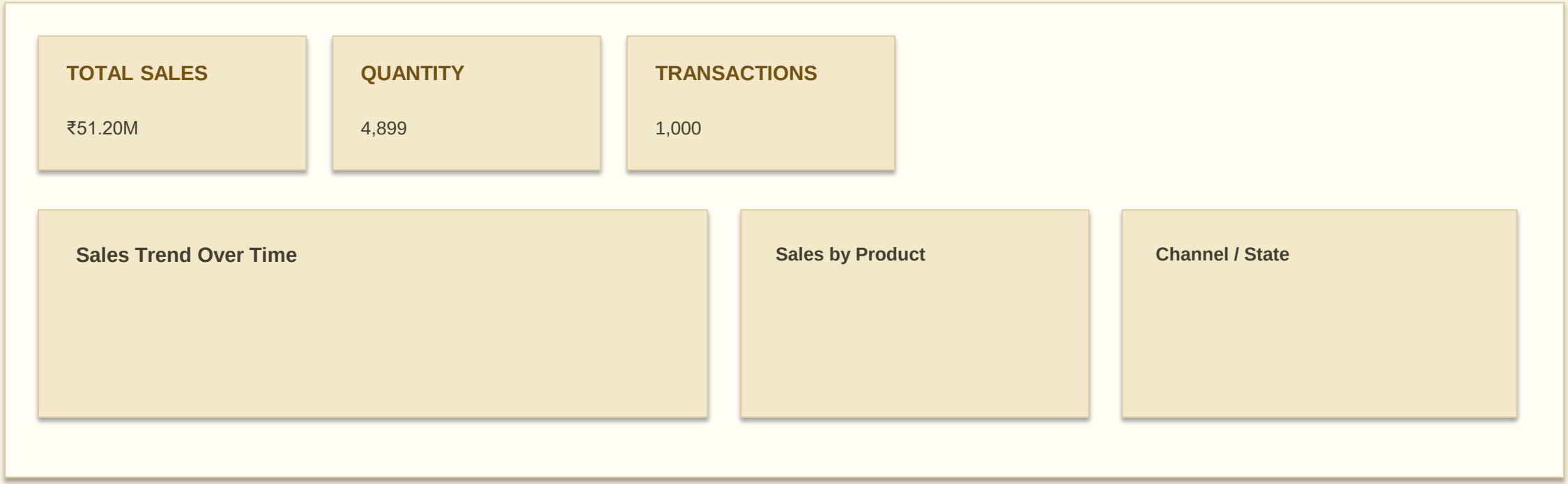
CUSTOMER COVERAGE

996 unique customer names are reported in the notebook summary.

The uploaded SQL also uses DENSE_RANK() to rank products by sales; its comment mentions customers, but the query groups by product.

Combine the analysis into one interactive report

Suggested dashboard layout based on fields in the uploaded dataset.



Recommended slicers: Date, Product, State, Sales Channel and Payment Type.

PROJECT TAKEAWAYS

From raw data to business insights

Python was used for exploration and data checks; MySQL stores and queries the salesdata table; Power BI presents the key metrics and comparisons.

DATA QUALITY

1,000 rows; notebook reports no nulls and no duplicated rows.

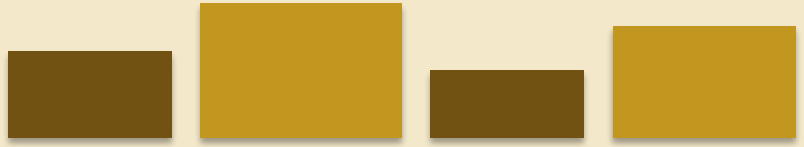
BUSINESS ANALYSIS

Revenue, product, channel, state and customer queries are defined in SQL.

NEXT STEP

Validate Power BI visuals and add verified product/channel/state rankings.

END OF PRESENTATION



Thank You!

Tech Products Sales Analysis

Python • MySQL • Power BI